

25.08.2020

### Kit components

| Product code                        | Description                                                  |
|-------------------------------------|--------------------------------------------------------------|
| <b>1662700</b><br><b>1662700EDU</b> | <b>Comparative Proteomics Kit I: Protein Profiler Module</b> |

#### Components:

|         |                                               |
|---------|-----------------------------------------------|
| 1610375 | Precision Plus Protein Kaleidoscope Standards |
| 1660010 | Actin / Myosin Standard                       |
| 1610732 | 10X Tris/Glycine/SDS Buffer                   |
| 1610737 | 2X Laemmli Sample Buffer                      |
| 1610786 | Bio-Safe Coomassie Stain                      |
| 1610610 | Dithiothreitol (DTT)                          |

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

### SECTION 1: Identification

- **1.1 Product identifier**
- **Trade name:** *Precision Plus Protein Kaleidoscope Standards*
- **Article number:** 1610375, 1610375S, 1610375EDU, 1610395, 10022172
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Laboratories Pte Ltd.  
27 International Business Park 01-02  
Singapore 609924
- **Information department:** sales.singapore@bio-rad.com
- **1.4 Emergency telephone number:** 65-6415-3188

phone: 65-6415-3188 fax: 65-6415-3189

### SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**  
Acute Tox. 5 H333 May be harmful if inhaled.  
Skin Corr. 3 H316 Causes mild skin irritation.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008**  
The product is classified and labelled according to the CLP regulation.
- **Hazard pictograms** Not Applicable
- **Signal word** Warning
- **Hazard statements**  
H333 May be harmful if inhaled.  
H316 Causes mild skin irritation.
- **Precautionary statements**  
P304+P312 IF INHALED: Call a POISON CENTER/doctor if you feel unwell.  
P332+P313 If skin irritation occurs: Get medical advice/attention.
- **2.3 Other hazards**
- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

- **3.2 Chemical characterisation: Mixtures**
- **Description:** Mixture of the substances listed below with nonhazardous additions.

· **Dangerous components:**

|                                    |                         |                                                                                                                           |        |
|------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|
| CAS: 56-81-5<br>EINECS: 200-289-5  | glycerol                | Acute Tox. 5, H303; Acute Tox. 5, H333                                                                                    | 20-35% |
| CAS: 151-21-3<br>EINECS: 205-788-1 | sodium dodecyl sulphate | Flam. Sol. 2, H228<br>Acute Tox. 3, H301; Acute Tox. 3, H311<br>Skin Irrit. 2, H315; Eye Irrit. 2A, H319; STOT SE 3, H335 | 1-2.5% |

(Contd. on page 2)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

**Trade name: Precision Plus Protein Kaleidoscope Standards**

(Contd. of page 1)

· **Additional information** For the wording of the listed hazard phrases refer to section 16.

### SECTION 4: First-aid measures

- **4.1 Description of first aid measures**
- **General information** No special measures required.
- **After inhalation** Supply fresh air; consult doctor in case of complaints.
- **After skin contact** Generally the product does not irritate the skin.
- **After eye contact** Rinse opened eye for several minutes under running water.
- **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.
- **4.2 Most important symptoms and effects, both acute and delayed** No further relevant information available.
- **4.3 Indication of any immediate medical attention and special treatment needed**  
No further relevant information available.

### SECTION 5: Fire-fighting measures

- **5.1 Extinguishing media**
- **Suitable extinguishing agents** Use fire fighting measures that suit the environment.
- **5.2 Special hazards arising from the substance or mixture** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** No special measures required.

### SECTION 6: Accidental release measures

- **6.1 Personal precautions, protective equipment and emergency procedures** Wear protective clothing.
- **6.2 Environmental precautions:**  
Dilute with plenty of water.  
Do not allow to enter sewers/ surface or ground water.
- **6.3 Methods and material for containment and cleaning up:**  
Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).
- **6.4 Reference to other sections**  
No dangerous substances are released.  
See Section 7 for information on safe handling  
See Section 8 for information on personal protection equipment.  
See Section 13 for disposal information.

### SECTION 7: Handling and storage

- **7.1 Precautions for safe handling** No special measures required.
- **Information about protection against explosions and fires:** No special measures required.
- **7.2 Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and receptacles:** According to product specification
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** None.

(Contd. on page 3)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

Trade name: Precision Plus Protein Kaleidoscope Standards

(Contd. of page 2)

· **7.3 Specific end use(s)** No further relevant information available.

### SECTION 8: Exposure controls/personal protection

· **Additional information about design of technical systems:** No further data; see item 7.· **8.1 Control parameters**· **Components with limit values that require monitoring at the workplace:****56-81-5 glycerol**WSH Long-term value: 10 mg/m<sup>3</sup>· **Additional information:** The lists that were valid during the creation were used as basis.· **8.2 Exposure controls**· **Personal protective equipment**· **General protective and hygienic measures** Wash hands before breaks and at the end of work.· **Breathing equipment:** Not necessary if room is well-ventilated.· **Protection of hands:** Protective gloves.· **Material of gloves** Synthetic gloves· **Penetration time of glove material**

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

· **Eye protection:** Safety glasses

### SECTION 9: Physical and chemical properties

· **9.1 Information on basic physical and chemical properties**· **General Information**· **Appearance:**

Form: Liquid

Colour: Violet

· **Odour:** Rotten· **Odour threshold:** Not determined.· **pH-value at 20 °C:** 6.8· **Change in condition**

Melting point/freezing point: undetermined

Initial boiling point and boiling range: 100 °C

· **Flash point:** > 100 °C· **Flammability (solid, gaseous)** Not applicable.· **Ignition temperature:** 400 °C· **Decomposition temperature:** Not determined.· **Self igniting:** Product is not selfigniting.· **Explosive properties:** Not determined.

(Contd. on page 4)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

Trade name: Precision Plus Protein Kaleidoscope Standards

(Contd. of page 3)

|                                                  |                                            |
|--------------------------------------------------|--------------------------------------------|
| · <b>Explosion limits:</b>                       |                                            |
| Lower:                                           | 0.9 Vol %                                  |
| Upper:                                           | Not determined.                            |
| · <b>Vapour pressure at 20 °C:</b>               | 23 hPa                                     |
| · <b>Density:</b>                                | Not determined                             |
| · <b>Relative density</b>                        | Not determined.                            |
| · <b>Vapour density</b>                          | Not determined.                            |
| · <b>Evaporation rate</b>                        | Not determined.                            |
| · <b>Solubility in / Miscibility with Water:</b> | Fully miscible                             |
| · <b>Partition coefficient: n-octanol/water:</b> | Not determined.                            |
| · <b>Viscosity:</b>                              |                                            |
| dynamic:                                         | Not determined.                            |
| kinematic:                                       | Not determined.                            |
| · <b>Solvent content:</b>                        |                                            |
| Organic solvents:                                | 30.0 %                                     |
| Water:                                           | 66.2 %                                     |
| Solids content:                                  | 3.8 %                                      |
| · <b>9.2 Other information</b>                   | No further relevant information available. |

### SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

### SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**
- **Acute toxicity**  
May be harmful if inhaled.

· **LD/LC50 values that are relevant for classification:**

#### 56-81-5 glycerol

|        |      |                                           |
|--------|------|-------------------------------------------|
| Oral   | LD50 | 4,090 mg/kg (mouse)<br>12,600 mg/kg (rat) |
| Dermal | LD50 | 10,000 mg/kg (Rabbit)                     |

(Contd. on page 5)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

**Trade name: Precision Plus Protein Kaleidoscope Standards**

(Contd. of page 4)

|                                         |          |                    |
|-----------------------------------------|----------|--------------------|
| Inhalative                              | LC50/4 h | >30 mg/l (Rat)     |
| <b>151-21-3 sodium dodecyl sulphate</b> |          |                    |
| Oral                                    | LD50     | 1,288 mg/kg (rat)  |
| Dermal                                  | LD50     | 580 mg/kg (Rabbit) |

- **Primary irritant effect:**
- **Skin corrosion/irritation**  
Causes mild skin irritation.
- **Serious eye damage/irritation** Based on available data, the classification criteria are not met.
- **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.
- **CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)**
- **Germ cell mutagenicity** Based on available data, the classification criteria are not met.
- **Carcinogenicity** Based on available data, the classification criteria are not met.
- **Reproductive toxicity** Based on available data, the classification criteria are not met.
- **STOT-single exposure** Based on available data, the classification criteria are not met.
- **STOT-repeated exposure** Based on available data, the classification criteria are not met.
- **Aspiration hazard** Based on available data, the classification criteria are not met.

### SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability** No further relevant information available.
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **Additional ecological information:**
- **General notes:**  
Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.  
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

### SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation**  
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.  
Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.
- **Recommended cleansing agent:** Water, if necessary with cleansing agents.

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(Contd. on page 6)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

Trade name: Precision Plus Protein Kaleidoscope Standards

(Contd. of page 5)

### SECTION 14: Transport information

|                                                                                  |                                                      |
|----------------------------------------------------------------------------------|------------------------------------------------------|
| · <b>14.1 UN-Number</b><br>· ADR, ADN, IMDG, IATA                                | Not Reguated                                         |
| · <b>14.2 UN proper shipping name</b><br>· ADR                                   | Not Reguated<br>Not Reguated                         |
| · ADN, IMDG, IATA                                                                | Not Reguated                                         |
| · <b>14.3 Transport hazard class(es)</b><br>· ADR, ADN, IMDG, IATA<br>· Class    | Not Reguated                                         |
| · <b>14.4 Packing group</b><br>· ADR, IMDG, IATA                                 | Not Reguated                                         |
| · <b>14.5 Environmental hazards:</b><br>· Marine pollutant:                      | No                                                   |
| · <b>14.6 Special precautions for user</b>                                       | Not applicable.                                      |
| · <b>14.7 Transport in bulk according to Annex II of Marpol and the IBC Code</b> | Not applicable.                                      |
| · <b>Transport/Additional information:</b>                                       | Not dangerous according to the above specifications. |
| · <b>UN "Model Regulation":</b>                                                  | Not Reguated                                         |

### SECTION 15: Regulatory information

#### · 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

##### · Poisons Act - Schedule 1

None of the ingredients is listed.

##### · Poisons Act - Schedule 2, Group II

None of the ingredients is listed.

##### · Health Products Act - First Schedule - Psychotropic Substances

None of the ingredients is listed.

##### · Directive 2012/18/EU

##### · Named dangerous substances - ANNEX I None of the ingredients is listed.

##### · National regulations

##### · Technical instructions (air):

| Class | Share in % |
|-------|------------|
| NK    | 20-35      |

##### · Water hazard class: Water hazard class 1 (Self-assessment): slightly hazardous for water.

(Contd. on page 7)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 25.07.2020

**Trade name: Precision Plus Protein Kaleidoscope Standards**

(Contd. of page 6)

· **15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

### SECTION 16: Other information

*This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.*

· **Department issuing SDS:** Environmental Health and Safety.

· **Contact:**

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

· **Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Flam. Sol. 2: Flammable solids – Category 2

Acute Tox. 3: Acute toxicity - oral – Category 3

Acute Tox. 5: Acute toxicity - oral – Category 5

Skin Irrit. 2: Skin corrosion/irritation – Category 2

Skin Corr. 3: Skin corrosion/irritation – Category 3

Eye Irrit. 2A: Serious eye damage/eye irritation – Category 2A

STOT SE 3: Specific target organ toxicity (single exposure) – Category 3

· **\* Data compared to the previous version altered.**



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

### SECTION 1: Identification

- **1.1 Product identifier**
- **Trade name:** Actin / Myosin Standard
- **Article number:** 1660010, 1660010EDU, 9703865
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Laboratories Pte Ltd.  
27 International Business Park 01-02  
Singapore 609924
- phone: 65-6415-3188 fax: 65-6415-3189
- **Information department:** sales.singapore@bio-rad.com
- **1.4 Emergency telephone number:** 65-6415-3188

### SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**  
Acute Tox. 5 H303 May be harmful if swallowed.  
Skin Irrit. 2 H315 Causes skin irritation.  
Eye Irrit. 2A H319 Causes serious eye irritation.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008**  
The product is classified and labelled according to the CLP regulation.
- **Hazard pictograms**



GHS07

- **Signal word** Warning
- **Hazard-determining components of labelling:**  
Dithiothreitol
- **Hazard statements**  
H303 May be harmful if swallowed.  
H315 Causes skin irritation.  
H319 Causes serious eye irritation.
- **Precautionary statements**  
P280 Wear protective gloves / eye protection / face protection.  
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
P312 Call a POISON CENTER/doctor if you feel unwell.  
P321 Specific treatment (see on this label).  
P362+P364 Take off contaminated clothing and wash it before reuse.  
P332+P313 If skin irritation occurs: Get medical advice/attention.

(Contd. on page 2)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: Actin / Myosin Standard

(Contd. of page 1)

- **2.3 Other hazards**
- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

- **3.2 Chemical characterisation: Mixtures**
- **Description:** Mixture of the substances listed below with nonhazardous additions.

- **Dangerous components:**

|                   |                                                                               |         |
|-------------------|-------------------------------------------------------------------------------|---------|
| CAS: 57-50-1      | sucrose, pure                                                                 | 50-100% |
| EINECS: 200-334-9 | substance with a Community workplace exposure limit                           |         |
| CAS: 3483-12-3    | Dithiothreitol                                                                | 10-<20% |
| EINECS: 222-468-7 | Acute Tox. 4, H302; Skin Irrit. 2, H315; Eye Irrit. 2A, H319; STOT SE 3, H335 |         |

- **Additional information** For the wording of the listed hazard phrases refer to section 16.

### SECTION 4: First-aid measures

- **4.1 Description of first aid measures**
- **General information** Immediately remove any clothing soiled by the product.
- **After inhalation** Supply fresh air; consult doctor in case of complaints.
- **After skin contact**  
Generally the product does not irritate the skin.  
Immediately wash with water and soap and rinse thoroughly.  
Immediately rinse with water.
- **After eye contact** Rinse opened eye for several minutes under running water. If symptoms persist, consult a doctor.
- **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.
- **4.2 Most important symptoms and effects, both acute and delayed** No further relevant information available.
- **4.3 Indication of any immediate medical attention and special treatment needed**  
No further relevant information available.

### SECTION 5: Fire-fighting measures

- **5.1 Extinguishing media**
- **Suitable extinguishing agents** Use fire fighting measures that suit the environment.
- **5.2 Special hazards arising from the substance or mixture** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** Wear self-contained respiratory protective device.

### SECTION 6: Accidental release measures

- **6.1 Personal precautions, protective equipment and emergency procedures** Wear protective clothing.
- **6.2 Environmental precautions:** No special measures required.
- **6.3 Methods and material for containment and cleaning up:** Pick up mechanically.

(Contd. on page 3)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: Actin / Myosin Standard

(Contd. of page 2)

### 6.4 Reference to other sections

- No dangerous substances are released.
- See Section 7 for information on safe handling
- See Section 8 for information on personal protection equipment.
- See Section 13 for disposal information.

## SECTION 7: Handling and storage

- 7.1 Precautions for safe handling No special measures required.
- Information about protection against explosions and fires: No special measures required.
- 7.2 Conditions for safe storage, including any incompatibilities
- Storage
- Requirements to be met by storerooms and receptacles: According to product specification
- Information about storage in one common storage facility: Not required.
- Further information about storage conditions: Keep receptacle tightly sealed.
- 7.3 Specific end use(s) No further relevant information available.

## SECTION 8: Exposure controls/personal protection

- Additional information about design of technical systems: No further data; see item 7.

### 8.1 Control parameters

- Components with limit values that require monitoring at the workplace:

#### 57-50-1 sucrose, pure

|     |                                       |
|-----|---------------------------------------|
| WSH | Long-term value: 10 mg/m <sup>3</sup> |
|-----|---------------------------------------|

- Additional information: The lists that were valid during the creation were used as basis.

### 8.2 Exposure controls

#### Personal protective equipment

#### General protective and hygienic measures

- Keep away from foodstuffs, beverages and feed.
- Immediately remove all soiled and contaminated clothing
- Wash hands before breaks and at the end of work.
- Avoid contact with the eyes.
- Avoid contact with the eyes and skin.

#### Protection of hands: Protective gloves.

#### Material of gloves Synthetic gloves

#### Penetration time of glove material

- The exact break trough time has to be found out by the manufacturer of the protective gloves and has to be observed.

#### Eye protection:

- Safety glasses
- Tightly sealed goggles.

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(Contd. on page 4)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: Actin / Myosin Standard

(Contd. of page 3)

### SECTION 9: Physical and chemical properties

#### · 9.1 Information on basic physical and chemical properties

##### · General Information

##### · Appearance:

|                  |                 |
|------------------|-----------------|
| Form:            | Solid.          |
| Colour:          | White           |
| Odour:           | Odourless       |
| Odour threshold: | Not determined. |

· pH-value: Not applicable.

##### · Change in condition

|                                          |              |
|------------------------------------------|--------------|
| Melting point/freezing point:            | undetermined |
| Initial boiling point and boiling range: | 1,500 °C     |

· Flash point: > 100 °C

· Flammability (solid, gaseous) Not determined.

· Decomposition temperature: Not determined.

· Self igniting: Product is not selfigniting.

· Explosive properties: Not determined.

##### · Explosion limits:

|        |                 |
|--------|-----------------|
| Lower: | Not determined. |
| Upper: | Not determined. |

· Vapour pressure: Not applicable.

|                  |                 |
|------------------|-----------------|
| Density:         | Not determined  |
| Relative density | Not determined. |
| Vapour density   | Not applicable. |
| Evaporation rate | Not applicable. |

##### · Solubility in / Miscibility with

|        |         |
|--------|---------|
| Water: | Soluble |
|--------|---------|

· Partition coefficient: n-octanol/water: Not determined.

##### · Viscosity:

|            |                 |
|------------|-----------------|
| dynamic:   | Not applicable. |
| kinematic: | Not applicable. |

##### · Solvent content:

|                 |        |
|-----------------|--------|
| Solids content: | 78.8 % |
|-----------------|--------|

· 9.2 Other information No further relevant information available.

### SECTION 10: Stability and reactivity

· 10.1 Reactivity No further relevant information available.

(Contd. on page 5)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: Actin / Myosin Standard

(Contd. of page 4)

- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

### SECTION 11: Toxicological information

#### · 11.1 Information on toxicological effects

##### · Acute toxicity

May be harmful if swallowed.

#### · LD/LC50 values that are relevant for classification:

##### **3483-12-3 Dithiothreitol**

|      |      |                 |
|------|------|-----------------|
| Oral | LD50 | 400 mg/kg (rat) |
|------|------|-----------------|

#### · Primary irritant effect:

##### · Skin corrosion/irritation

Causes skin irritation.

##### · Serious eye damage/irritation

Causes serious eye irritation.

##### · Respiratory or skin sensitisation

 Based on available data, the classification criteria are not met.

##### · CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)

##### · Germ cell mutagenicity

 Based on available data, the classification criteria are not met.

##### · Carcinogenicity

 Based on available data, the classification criteria are not met.

##### · Reproductive toxicity

 Based on available data, the classification criteria are not met.

##### · STOT-single exposure

 Based on available data, the classification criteria are not met.

##### · STOT-repeated exposure

 Based on available data, the classification criteria are not met.

##### · Aspiration hazard

 Based on available data, the classification criteria are not met.

### SECTION 12: Ecological information

#### · 12.1 Toxicity

##### · Aquatic toxicity:

 No further relevant information available.

##### · 12.2 Persistence and degradability

 No further relevant information available.

##### · 12.3 Bioaccumulative potential

 No further relevant information available.

##### · 12.4 Mobility in soil

 No further relevant information available.

#### · Additional ecological information:

##### · General notes:

 Generally not hazardous for water.

##### · 12.5 Results of PBT and vPvB assessment

##### · PBT:

 Not applicable.

##### · vPvB:

 Not applicable.

##### · 12.6 Other adverse effects

 No further relevant information available.

SG

(Contd. on page 6)

## Safety data sheet

### according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: Actin / Myosin Standard

(Contd. of page 5)

### SECTION 13: Disposal considerations

#### · 13.1 Waste treatment methods

##### · Recommendation

Must not be disposed of together with household garbage. Do not allow product to reach sewage system.  
Dispose of waste in accordance to applicable national, regional, or local regulations.

##### · Uncleaned packagings:

· Recommendation: Disposal must be made according to official regulations.

· Recommended cleansing agent: Water, if necessary with cleansing agents.

### SECTION 14: Transport information

#### · 14.1 UN-Number

· ADR, ADN, IMDG, IATA

Not Reguated

#### · 14.2 UN proper shipping name

· ADR

Not Reguated

Not Reguated

· ADN, IMDG, IATA

Not Reguated

#### · 14.3 Transport hazard class(es)

· ADR, ADN, IMDG, IATA

· Class

Not Reguated

#### · 14.4 Packing group

· ADR, IMDG, IATA

Not Reguated

#### · 14.5 Environmental hazards:

· Marine pollutant:

No

#### · 14.6 Special precautions for user

Not applicable.

#### · 14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable.

#### · Transport/Additional information:

Not dangerous according to the above specifications.

#### · UN "Model Regulation":

Not Reguated

### SECTION 15: Regulatory information

#### · 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

##### · Poisons Act - Schedule 1

None of the ingredients is listed.

##### · Poisons Act - Schedule 2, Group II

None of the ingredients is listed.

(Contd. on page 7)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: Actin / Myosin Standard

(Contd. of page 6)

**Health Products Act - First Schedule - Psychotropic Substances**

None of the ingredients is listed.

**Directive 2012/18/EU**
**Named dangerous substances - ANNEX I** None of the ingredients is listed.

**National regulations**
**Water hazard class:** Generally not hazardous for water.

**15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

### SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

**Department issuing SDS:** Environmental Health and Safety.

**Contact:**

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

**Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Acute Tox. 4: Acute toxicity - oral – Category 4

Acute Tox. 5: Acute toxicity - oral – Category 5

Skin Irrit. 2: Skin corrosion/irritation – Category 2

Eye Irrit. 2A: Serious eye damage/eye irritation – Category 2A

STOT SE 3: Specific target organ toxicity (single exposure) – Category 3

**\* Data compared to the previous version altered.**



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

### SECTION 1: Identification

- **1.1 Product identifier**
- **Trade name:** **10X Tris/Glycine/SDS Buffer**
- **Article number:** 1610732, 9701906, 1610772, 1610732EDU, 1610772EDU, 10021723
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Laboratories Pte Ltd.  
27 International Business Park 01-02  
Singapore 609924
- phone: 65-6415-3188 fax: 65-6415-3189
- **Information department:** sales.singapore@bio-rad.com
- **1.4 Emergency telephone number:** 65-6415-3188

### SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008** The product is not classified, according to the CLP regulation.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008** Not Applicable
- **Hazard pictograms** Not Applicable
- **Signal word** Not Applicable
- **Hazard statements** Not Applicable
- **2.3 Other hazards**
- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

- **3.2 Chemical characterisation: Mixtures**
- **Description:** Mixture of the substances listed below with nonhazardous additions.
- **Dangerous components:** Not applicable.
- **Additional information** For the wording of the listed hazard phrases refer to section 16.

### SECTION 4: First-aid measures

- **4.1 Description of first aid measures**
- **General information** No special measures required.
- **After inhalation** Supply fresh air; consult doctor in case of complaints.
- **After skin contact** Generally the product does not irritate the skin.
- **After eye contact** Rinse opened eye for several minutes under running water.
- **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.
- **4.2 Most important symptoms and effects, both acute and delayed** No further relevant information available.

(Contd. on page 2)



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: 10X Tris/Glycine/SDS Buffer

(Contd. of page 1)

- **4.3 Indication of any immediate medical attention and special treatment needed**  
No further relevant information available.

### SECTION 5: Fire-fighting measures

- **5.1 Extinguishing media**
- **Suitable extinguishing agents** Use fire fighting measures that suit the environment.
- **5.2 Special hazards arising from the substance or mixture** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** No special measures required.

### SECTION 6: Accidental release measures

- **6.1 Personal precautions, protective equipment and emergency procedures** Wear protective clothing.
- **6.2 Environmental precautions:**  
Dilute with plenty of water.  
Do not allow to enter sewers/ surface or ground water.
- **6.3 Methods and material for containment and cleaning up:**  
Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).
- **6.4 Reference to other sections**  
No dangerous substances are released.  
See Section 7 for information on safe handling  
See Section 8 for information on personal protection equipment.  
See Section 13 for disposal information.

### SECTION 7: Handling and storage

- **7.1 Precautions for safe handling** No special measures required.
- **Information about protection against explosions and fires:** No special measures required.
- **7.2 Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and receptacles:** According to product specification
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** None.
- **7.3 Specific end use(s)** No further relevant information available.

### SECTION 8: Exposure controls/personal protection

- **Additional information about design of technical systems:** No further data; see item 7.
- **8.1 Control parameters**
- **Components with limit values that require monitoring at the workplace:**  
The product does not contain any relevant quantities of materials with critical values that have to be monitored at the workplace.
- **Additional information:** The lists that were valid during the creation were used as basis.

(Contd. on page 3)

## Safety data sheet

### according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: 10X Tris/Glycine/SDS Buffer

(Contd. of page 2)

- **8.2 Exposure controls**
- **Personal protective equipment**
- **General protective and hygienic measures**  
The usual precautionary measures should be adhered to when handling chemicals.
- **Breathing equipment:** Not required.
- **Protection of hands:** Protective gloves.
- **Material of gloves:** Synthetic gloves
- **Penetration time of glove material**  
The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.
- **Eye protection:** Safety glasses

## SECTION 9: Physical and chemical properties

### · 9.1 Information on basic physical and chemical properties

#### · General Information

##### · Appearance:

|                  |                 |
|------------------|-----------------|
| Form:            | Fluid           |
| Colour:          | Light yellow    |
| Odour:           | Odourless       |
| Odour threshold: | Not determined. |

|                      |     |
|----------------------|-----|
| · pH-value at 20 °C: | 8.3 |
|----------------------|-----|

#### · Change in condition

|                                          |              |
|------------------------------------------|--------------|
| Melting point/freezing point:            | undetermined |
| Initial boiling point and boiling range: | 100 °C       |

|                |                                   |
|----------------|-----------------------------------|
| · Flash point: | Not applicable<br>Not determined. |
|----------------|-----------------------------------|

|                                 |                 |
|---------------------------------|-----------------|
| · Flammability (solid, gaseous) | Not applicable. |
|---------------------------------|-----------------|

|                              |                 |
|------------------------------|-----------------|
| · Decomposition temperature: | Not determined. |
|------------------------------|-----------------|

|                  |                              |
|------------------|------------------------------|
| · Self igniting: | Product is not selfigniting. |
|------------------|------------------------------|

|                         |                 |
|-------------------------|-----------------|
| · Explosive properties: | Not determined. |
|-------------------------|-----------------|

#### · Explosion limits:

|        |                 |
|--------|-----------------|
| Lower: | Not determined. |
| Upper: | Not determined. |

|                             |        |
|-----------------------------|--------|
| · Vapour pressure at 20 °C: | 23 hPa |
|-----------------------------|--------|

|                     |                           |
|---------------------|---------------------------|
| · Density at 20 °C: | 0.98915 g/cm <sup>3</sup> |
|---------------------|---------------------------|

|                    |                 |
|--------------------|-----------------|
| · Relative density | Not determined. |
|--------------------|-----------------|

|                  |                 |
|------------------|-----------------|
| · Vapour density | Not determined. |
|------------------|-----------------|

|                    |                 |
|--------------------|-----------------|
| · Evaporation rate | Not determined. |
|--------------------|-----------------|

#### · Solubility in / Miscibility with

|        |                |
|--------|----------------|
| Water: | Fully miscible |
|--------|----------------|

(Contd. on page 4)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: 10X Tris/Glycine/SDS Buffer

(Contd. of page 3)

- **Partition coefficient: n-octanol/water:** Not determined.
- **Viscosity:**  
   dynamic: Not determined.  
   kinematic: Not determined.
- **Solvent content:**  
   Water: 81.4 %
- **Solids content:** 18.6 %
- **9.2 Other information** No further relevant information available.

### SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

### SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**
- **Acute toxicity** Based on available data, the classification criteria are not met.

#### · LD/LC50 values that are relevant for classification:

##### 56-40-6 glycine

Oral LD50 7,930 mg/kg (rat)

Dermal LD50 5,200 mg/kg (rat)

##### 77-86-1 Tris(hydroxymethyl)aminomethane

Oral LD50 5,900 mg/kg (Rat)

Dermal LD50 &gt;5,000 mg/kg (Rat)

- **Primary irritant effect:**
- **Skin corrosion/irritation** Based on available data, the classification criteria are not met.
- **Serious eye damage/irritation** Based on available data, the classification criteria are not met.
- **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.
- **CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)**
- **Germ cell mutagenicity** Based on available data, the classification criteria are not met.
- **Carcinogenicity** Based on available data, the classification criteria are not met.
- **Reproductive toxicity** Based on available data, the classification criteria are not met.
- **STOT-single exposure** Based on available data, the classification criteria are not met.
- **STOT-repeated exposure** Based on available data, the classification criteria are not met.

(Contd. on page 5)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: 10X Tris/Glycine/SDS Buffer

(Contd. of page 4)

· **Aspiration hazard** Based on available data, the classification criteria are not met.

### SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability** No further relevant information available.
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **Additional ecological information:**
- **General notes:**  
Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.  
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

### SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation**  
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.  
Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.
- **Recommended cleansing agent:** Water, if necessary with cleansing agents.

### SECTION 14: Transport information

- |                                                                                                                                                           |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <ul style="list-style-type: none"> <li>· <b>14.1 UN-Number</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> </ul>                                          | Not Reguated                                 |
| <ul style="list-style-type: none"> <li>· <b>14.2 UN proper shipping name</b></li> <li>· <b>ADR</b></li> <li>· <b>ADN, IMDG, IATA</b></li> </ul>           | Not Reguated<br>Not Reguated<br>Not Reguated |
| <ul style="list-style-type: none"> <li>· <b>14.3 Transport hazard class(es)</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> <li>· <b>Class</b></li> </ul> | Not Reguated                                 |
| <ul style="list-style-type: none"> <li>· <b>14.4 Packing group</b></li> <li>· <b>ADR, IMDG, IATA</b></li> </ul>                                           | Not Reguated                                 |

(Contd. on page 6)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 24.07.2020

Trade name: 10X Tris/Glycine/SDS Buffer

(Contd. of page 5)

|                                                                                  |                                                      |
|----------------------------------------------------------------------------------|------------------------------------------------------|
| · <b>14.5 Environmental hazards:</b>                                             |                                                      |
| · <b>Marine pollutant:</b>                                                       | No                                                   |
| · <b>14.6 Special precautions for user</b>                                       | Not applicable.                                      |
| · <b>14.7 Transport in bulk according to Annex II of Marpol and the IBC Code</b> | Not applicable.                                      |
| · <b>Transport/Additional information:</b>                                       | Not dangerous according to the above specifications. |
| · <b>UN "Model Regulation":</b>                                                  | Not Reguated                                         |

### SECTION 15: Regulatory information

#### · 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

##### · Poisons Act - Schedule 1

None of the ingredients is listed.

##### · Poisons Act - Schedule 2, Group II

None of the ingredients is listed.

##### · Health Products Act - First Schedule - Psychotropic Substances

None of the ingredients is listed.

##### · Directive 2012/18/EU

##### · Named dangerous substances - ANNEX I None of the ingredients is listed.

##### · National regulations

##### · Water hazard class: Water hazard class 1 (Self-assessment): slightly hazardous for water.

##### · 15.2 Chemical safety assessment: A Chemical Safety Assessment has not been carried out.

### SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

#### · Department issuing SDS: Environmental Health and Safety.

#### · Contact:

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

#### · Abbreviations and acronyms:

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

(Contd. on page 7)

**Safety data sheet**  
**according SS 586 : Part 3**

Printing date 25.08.2020

Revision: 24.07.2020

**Trade name: 10X Tris/Glycine/SDS Buffer**

(Contd. of page 6)

*LD50: Lethal dose, 50 percent**PBT: Persistent, Bioaccumulative and Toxic**vPvB: very Persistent and very Bioaccumulative***\* Data compared to the previous version altered.**

SG

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

### SECTION 1: Identification

- **1.1 Product identifier**
- **Trade name:** **2X Laemmli Sample Buffer**
- **Article number:** 1610737, 1610737EDU, 1610737XTU
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Laboratories Pte Ltd.  
27 International Business Park 01-02  
Singapore 609924
- **Information department:** sales.singapore@bio-rad.com
- **1.4 Emergency telephone number:** 65-6415-3188

phone: 65-6415-3188 fax: 65-6415-3189

### SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**  
Acute Tox. 5 H333 May be harmful if inhaled.  
Skin Corr. 3 H316 Causes mild skin irritation.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008**  
The product is classified and labelled according to the CLP regulation.
- **Hazard pictograms** Not Applicable
- **Signal word** Warning
- **Hazard statements**  
H333 May be harmful if inhaled.  
H316 Causes mild skin irritation.
- **Precautionary statements**  
P304+P312 IF INHALED: Call a POISON CENTER/doctor if you feel unwell.  
P332+P313 If skin irritation occurs: Get medical advice/attention.
- **2.3 Other hazards**
- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

- **3.2 Chemical characterisation: Mixtures**
- **Description:** Mixture of the substances listed below with nonhazardous additions.

· **Dangerous components:**

|                                    |                         |                                                                                                                           |        |
|------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|
| CAS: 56-81-5<br>EINECS: 200-289-5  | glycerol                | Acute Tox. 5, H303; Acute Tox. 5, H333                                                                                    | 20-35% |
| CAS: 151-21-3<br>EINECS: 205-788-1 | sodium dodecyl sulphate | Flam. Sol. 2, H228<br>Acute Tox. 3, H301; Acute Tox. 3, H311<br>Skin Irrit. 2, H315; Eye Irrit. 2A, H319; STOT SE 3, H335 | 1-2.5% |

(Contd. on page 2)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

**Trade name:** 2X Laemmli Sample Buffer

(Contd. of page 1)

· **Additional information** For the wording of the listed hazard phrases refer to section 16.

### SECTION 4: First-aid measures

- **4.1 Description of first aid measures**
- **General information** No special measures required.
- **After inhalation** Supply fresh air; consult doctor in case of complaints.
- **After skin contact** Generally the product does not irritate the skin.
- **After eye contact** Rinse opened eye for several minutes under running water.
- **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.
- **4.2 Most important symptoms and effects, both acute and delayed** No further relevant information available.
- **4.3 Indication of any immediate medical attention and special treatment needed**  
No further relevant information available.

### SECTION 5: Fire-fighting measures

- **5.1 Extinguishing media**
- **Suitable extinguishing agents** Use fire fighting measures that suit the environment.
- **5.2 Special hazards arising from the substance or mixture** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** No special measures required.

### SECTION 6: Accidental release measures

- **6.1 Personal precautions, protective equipment and emergency procedures** Wear protective clothing.
- **6.2 Environmental precautions:**  
Dilute with plenty of water.  
Do not allow to enter sewers/ surface or ground water.
- **6.3 Methods and material for containment and cleaning up:**  
Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).
- **6.4 Reference to other sections**  
No dangerous substances are released.  
See Section 7 for information on safe handling  
See Section 8 for information on personal protection equipment.  
See Section 13 for disposal information.

### SECTION 7: Handling and storage

- **7.1 Precautions for safe handling** No special measures required.
- **Information about protection against explosions and fires:** No special measures required.
- **7.2 Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and receptacles:** According to product specification
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** None.

(Contd. on page 3)



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

Trade name: 2X Laemmli Sample Buffer

(Contd. of page 2)

· **7.3 Specific end use(s)** No further relevant information available.

### SECTION 8: Exposure controls/personal protection

· **Additional information about design of technical systems:** No further data; see item 7.

#### · 8.1 Control parameters

· **Components with limit values that require monitoring at the workplace:**

**56-81-5 glycerol**

WSH Long-term value: 10 mg/m<sup>3</sup>

· **Additional information:** The lists that were valid during the creation were used as basis.

#### · 8.2 Exposure controls

· **Personal protective equipment**

· **General protective and hygienic measures**

The usual precautionary measures should be adhered to when handling chemicals.

· **Breathing equipment:** Not required.

· **Protection of hands:** Protective gloves.

· **Material of gloves** Synthetic gloves

· **Penetration time of glove material**

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

· **Eye protection:** Safety glasses

### SECTION 9: Physical and chemical properties

#### · 9.1 Information on basic physical and chemical properties

· **General Information**

· **Appearance:**

Form: Liquid

Colour: Light blue

· **Odour:** Odourless

· **Odour threshold:** Not determined.

· **pH-value at 20 °C:** 6.8

· **Change in condition**

Melting point/freezing point: undetermined

Initial boiling point and boiling range: 100 °C

· **Flash point:** > 100 °C

· **Flammability (solid, gaseous)** Not applicable.

· **Ignition temperature:** 400 °C

· **Decomposition temperature:** Not determined.

· **Self igniting:** Product is not selfigniting.

· **Explosive properties:** Not determined.

(Contd. on page 4)

## Safety data sheet

### according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

**Trade name: 2X Laemmli Sample Buffer**

(Contd. of page 3)

|                                                  |                                            |
|--------------------------------------------------|--------------------------------------------|
| · <b>Explosion limits:</b>                       |                                            |
| <b>Lower:</b>                                    | 0.9 Vol %                                  |
| <b>Upper:</b>                                    | Not determined.                            |
| · <b>Vapour pressure at 20 °C:</b>               | 23 hPa                                     |
| · <b>Density:</b>                                | Not determined                             |
| · <b>Relative density</b>                        | Not determined.                            |
| · <b>Vapour density</b>                          | Not determined.                            |
| · <b>Evaporation rate</b>                        | Not determined.                            |
| · <b>Solubility in / Miscibility with Water:</b> | Fully miscible                             |
| · <b>Partition coefficient: n-octanol/water:</b> | Not determined.                            |
| · <b>Viscosity:</b>                              |                                            |
| <b>dynamic:</b>                                  | Not determined.                            |
| <b>kinematic:</b>                                | Not determined.                            |
| · <b>Solvent content:</b>                        |                                            |
| <b>Organic solvents:</b>                         | 26.3 %                                     |
| <b>Water:</b>                                    | 70.8 %                                     |
| <b>Solids content:</b>                           | 2.9 %                                      |
| · <b>9.2 Other information</b>                   | No further relevant information available. |

## SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

## SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**
- **Acute toxicity**  
May be harmful if inhaled.

· **LD/LC50 values that are relevant for classification:**
**56-81-5 glycerol**

|        |      |                                           |
|--------|------|-------------------------------------------|
| Oral   | LD50 | 4,090 mg/kg (mouse)<br>12,600 mg/kg (rat) |
| Dermal | LD50 | 10,000 mg/kg (Rabbit)                     |

(Contd. on page 5)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

Trade name: 2X Laemmli Sample Buffer

(Contd. of page 4)

|                                         |          |                    |
|-----------------------------------------|----------|--------------------|
| Inhalative                              | LC50/4 h | >30 mg/l (Rat)     |
| <b>151-21-3 sodium dodecyl sulphate</b> |          |                    |
| Oral                                    | LD50     | 1,288 mg/kg (rat)  |
| Dermal                                  | LD50     | 580 mg/kg (Rabbit) |

- **Primary irritant effect:**
- **Skin corrosion/irritation**  
Causes mild skin irritation.
- **Serious eye damage/irritation** Based on available data, the classification criteria are not met.
- **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.
- **CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)**
- **Germ cell mutagenicity** Based on available data, the classification criteria are not met.
- **Carcinogenicity** Based on available data, the classification criteria are not met.
- **Reproductive toxicity** Based on available data, the classification criteria are not met.
- **STOT-single exposure** Based on available data, the classification criteria are not met.
- **STOT-repeated exposure** Based on available data, the classification criteria are not met.
- **Aspiration hazard** Based on available data, the classification criteria are not met.

### SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability** No further relevant information available.
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **Additional ecological information:**
- **General notes:**  
Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.  
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

### SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation**  
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.  
Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.
- **Recommended cleansing agent:** Water, if necessary with cleansing agents.

SG

(Contd. on page 6)

## Safety data sheet

### according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

Trade name: 2X Laemmli Sample Buffer

(Contd. of page 5)

#### SECTION 14: Transport information

|                                                                           |                                                      |
|---------------------------------------------------------------------------|------------------------------------------------------|
| · 14.1 UN-Number                                                          |                                                      |
| · ADR, ADN, IMDG, IATA                                                    | Not Reguated                                         |
| · 14.2 UN proper shipping name                                            |                                                      |
| · ADR                                                                     | Not Reguated                                         |
|                                                                           | Not Reguated                                         |
| · ADN, IMDG, IATA                                                         | Not Reguated                                         |
| · 14.3 Transport hazard class(es)                                         |                                                      |
| · ADR, ADN, IMDG, IATA                                                    |                                                      |
| · Class                                                                   | Not Reguated                                         |
| · 14.4 Packing group                                                      |                                                      |
| · ADR, IMDG, IATA                                                         | Not Reguated                                         |
| · 14.5 Environmental hazards:                                             |                                                      |
| · Marine pollutant:                                                       | No                                                   |
| · 14.6 Special precautions for user                                       | Not applicable.                                      |
| · 14.7 Transport in bulk according to Annex II of Marpol and the IBC Code | Not applicable.                                      |
| · Transport/Additional information:                                       | Not dangerous according to the above specifications. |
| · UN "Model Regulation":                                                  | Not Reguated                                         |

#### SECTION 15: Regulatory information

##### · 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

###### · Poisons Act - Schedule 1

None of the ingredients is listed.

###### · Poisons Act - Schedule 2, Group II

None of the ingredients is listed.

###### · Health Products Act - First Schedule - Psychotropic Substances

None of the ingredients is listed.

###### · Directive 2012/18/EU

###### · Named dangerous substances - ANNEX I None of the ingredients is listed.

###### · National regulations

###### · Technical instructions (air):

| Class | Share in % |
|-------|------------|
| NK    | 20-35      |

###### · Water hazard class: Water hazard class 1 (Self-assessment): slightly hazardous for water.

(Contd. on page 7)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 13.02.2020

Trade name: 2X Laemmli Sample Buffer

(Contd. of page 6)

· **15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

### SECTION 16: Other information

*This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.*

· **Department issuing SDS:** Environmental Health and Safety.

· **Contact:**

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

· **Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Flam. Sol. 2: Flammable solids – Category 2

Acute Tox. 3: Acute toxicity - oral – Category 3

Acute Tox. 5: Acute toxicity - oral – Category 5

Skin Irrit. 2: Skin corrosion/irritation – Category 2

Skin Corr. 3: Skin corrosion/irritation – Category 3

Eye Irrit. 2A: Serious eye damage/eye irritation – Category 2A

STOT SE 3: Specific target organ toxicity (single exposure) – Category 3

· **\* Data compared to the previous version altered.**

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

### SECTION 1: Identification

- **1.1 Product identifier**
- **Trade name:** **Bio-Safe Coomassie Stain**
- **Article number:** 1610786, 1610787, 1610786S, 1610787EDU, 9704607, 4307131, 4307045, 1610786EDU
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Laboratories Pte Ltd.  
27 International Business Park 01-02  
Singapore 609924  
phone: 65-6415-3188 fax: 65-6415-3189
- **Information department:** sales.singapore@bio-rad.com
- **1.4 Emergency telephone number:** 65-6415-3188

### SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**  
Acute Tox. 5 H333 May be harmful if inhaled.  
Skin Corr. 3 H316 Causes mild skin irritation.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008**  
The product is classified and labelled according to the CLP regulation.
- **Hazard pictograms** Not Applicable
- **Signal word** Warning
- **Hazard-determining components of labelling:**  
methanol
- **Hazard statements**  
H333 May be harmful if inhaled.  
H316 Causes mild skin irritation.
- **Precautionary statements**  
P304+P312 IF INHALED: Call a POISON CENTER/doctor if you feel unwell.  
P332+P313 If skin irritation occurs: Get medical advice/attention.
- **2.3 Other hazards**
- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

- **3.2 Chemical characterisation: Mixtures**
- **Description:** Mixture of the substances listed below with nonhazardous additions.

(Contd. on page 2)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

**Trade name: Bio-Safe Coomassie Stain**

(Contd. of page 1)

**· Dangerous components:**

|                                     |                 |                                                                                                     |        |
|-------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------|--------|
| CAS: 7664-38-2<br>EINECS: 231-633-2 | phosphoric acid | Skin Corr. 1B, H314                                                                                 | 2.5-5% |
| CAS: 67-56-1<br>EINECS: 200-659-6   | methanol        | Flam. Liq. 2, H225<br>Acute Tox. 3, H301; Acute Tox. 3, H311; Acute Tox. 3, H331<br>STOT SE 1, H370 | 0.1-1% |

**· Additional information** For the wording of the listed hazard phrases refer to section 16.**SECTION 4: First-aid measures**

- **4.1 Description of first aid measures**
- **General information** No special measures required.
- **After inhalation** Supply fresh air; consult doctor in case of complaints.
- **After skin contact** Generally the product does not irritate the skin.
- **After eye contact** Rinse opened eye for several minutes under running water.
- **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.
- **4.2 Most important symptoms and effects, both acute and delayed** No further relevant information available.
- **4.3 Indication of any immediate medical attention and special treatment needed**  
No further relevant information available.

**SECTION 5: Fire-fighting measures**

- **5.1 Extinguishing media**
- **Suitable extinguishing agents** Use fire fighting measures that suit the environment.
- **5.2 Special hazards arising from the substance or mixture** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** No special measures required.

**SECTION 6: Accidental release measures**

- **6.1 Personal precautions, protective equipment and emergency procedures** Wear protective clothing.
- **6.2 Environmental precautions:**  
Dilute with plenty of water.  
Do not allow to enter sewers/ surface or ground water.
- **6.3 Methods and material for containment and cleaning up:**  
Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).
- **6.4 Reference to other sections**  
No dangerous substances are released.  
See Section 7 for information on safe handling  
See Section 8 for information on personal protection equipment.  
See Section 13 for disposal information.

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(Contd. on page 3)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

Trade name: Bio-Safe Coomassie Stain

(Contd. of page 2)

### SECTION 7: Handling and storage

- **7.1 Precautions for safe handling** No special measures required.
- **Information about protection against explosions and fires:** No special measures required.
- **7.2 Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and receptacles:** According to product specification
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** None.
- **7.3 Specific end use(s)** No further relevant information available.

### SECTION 8: Exposure controls/personal protection

- **Additional information about design of technical systems:** No further data; see item 7.

#### · 8.1 Control parameters

- **Components with limit values that require monitoring at the workplace:**

#### 7664-38-2 phosphoric acid

|     |                                                                               |
|-----|-------------------------------------------------------------------------------|
| PEL | Short-term value: 3 mg/m <sup>3</sup><br>Long-term value: 1 mg/m <sup>3</sup> |
| WSH | Short-term value: 3 mg/m <sup>3</sup><br>Long-term value: 1 mg/m <sup>3</sup> |

#### 67-56-1 methanol

|     |                                                                                                       |
|-----|-------------------------------------------------------------------------------------------------------|
| PEL | Short-term value: 328 mg/m <sup>3</sup> , 250 ppm<br>Long-term value: 262 mg/m <sup>3</sup> , 200 ppm |
| WSH | Short-term value: 328 mg/m <sup>3</sup> , 250 ppm<br>Long-term value: 262 mg/m <sup>3</sup> , 200 ppm |

- **Additional information:** The lists that were valid during the creation were used as basis.
- **8.2 Exposure controls**
- **Personal protective equipment**
- **General protective and hygienic measures**  
The usual precautionary measures should be adhered to when handling chemicals.
- **Breathing equipment:** Not required.
- **Protection of hands:** Protective gloves.
- **Material of gloves** Synthetic gloves
- **Penetration time of glove material**  
The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.
- **Eye protection:** Safety glasses

SG

(Contd. on page 4)



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

Trade name: Bio-Safe Coomassie Stain

(Contd. of page 3)

### SECTION 9: Physical and chemical properties

#### · 9.1 Information on basic physical and chemical properties

##### · General Information

##### · Appearance:

Form: Liquid

Colour: Blue

· Odour: Acid

· Odour threshold: Not determined.

· pH-value: Not determined.

##### · Change in condition

Melting point/freezing point: undetermined

Initial boiling point and boiling range: 100 °C

· Flash point: Not applicable

· Flammability (solid, gaseous) Not applicable.

· Decomposition temperature: Not determined.

· Self igniting: Product is not selfigniting.

· Explosive properties: Not determined.

##### · Explosion limits:

Lower: Not determined.

Upper: Not determined.

· Vapour pressure at 20 °C: 23 hPa

· Density: Not determined

· Relative density Not determined.

· Vapour density Not determined.

· Evaporation rate Not determined.

##### · Solubility in / Miscibility with

Water: Fully miscible

· Partition coefficient: n-octanol/water: Not determined.

##### · Viscosity:

dynamic: Not determined.

kinematic: Not determined.

##### · Solvent content:

Organic solvents: 1.1 %

Water: 93.9 %

Solids content: 5.0 %

· 9.2 Other information No further relevant information available.

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

Trade name: Bio-Safe Coomassie Stain

(Contd. of page 4)

### SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

### SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**

- **Acute toxicity**  
May be harmful if inhaled.

- **LD/LC50 values that are relevant for classification:**

#### 67-56-1 methanol

|        |      |                    |
|--------|------|--------------------|
| Oral   | LD50 | 5,628 mg/kg (rat)  |
| Dermal | LD50 | 20,000 mg/kg (rat) |

- **Primary irritant effect:**
- **Skin corrosion/irritation**  
Causes mild skin irritation.
- **Serious eye damage/irritation** Based on available data, the classification criteria are not met.
- **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.
- **CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)**
- **Germ cell mutagenicity** Based on available data, the classification criteria are not met.
- **Carcinogenicity** Based on available data, the classification criteria are not met.
- **Reproductive toxicity** Based on available data, the classification criteria are not met.
- **STOT-single exposure** Based on available data, the classification criteria are not met.
- **STOT-repeated exposure** Based on available data, the classification criteria are not met.
- **Aspiration hazard** Based on available data, the classification criteria are not met.

### SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability** No further relevant information available.
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **Additional ecological information:**
- **General notes:**  
Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.  
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.

(Contd. on page 6)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

Trade name: Bio-Safe Coomassie Stain

(Contd. of page 5)

- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

### SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation**  
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.  
Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.
- **Recommended cleansing agent:** Water, if necessary with cleansing agents.

### SECTION 14: Transport information

- |                                                                                                                                                           |                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>· <b>14.1 UN-Number</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> </ul>                                          | <p>Not Reguated</p>                                         |
| <ul style="list-style-type: none"> <li>· <b>14.2 UN proper shipping name</b></li> <li>· <b>ADR</b></li> <li>· <b>ADN, IMDG, IATA</b></li> </ul>           | <p>Not Reguated<br/>Not Reguated<br/>Not Reguated</p>       |
| <ul style="list-style-type: none"> <li>· <b>14.3 Transport hazard class(es)</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> <li>· <b>Class</b></li> </ul> | <p>Not Reguated</p>                                         |
| <ul style="list-style-type: none"> <li>· <b>14.4 Packing group</b></li> <li>· <b>ADR, IMDG, IATA</b></li> </ul>                                           | <p>Not Reguated</p>                                         |
| <ul style="list-style-type: none"> <li>· <b>14.5 Environmental hazards:</b></li> <li>· <b>Marine pollutant:</b></li> </ul>                                | <p>No</p>                                                   |
| <ul style="list-style-type: none"> <li>· <b>14.6 Special precautions for user</b></li> </ul>                                                              | <p>Not applicable.</p>                                      |
| <ul style="list-style-type: none"> <li>· <b>14.7 Transport in bulk according to Annex II of Marpol and the IBC Code</b></li> </ul>                        | <p>Not applicable.</p>                                      |
| <ul style="list-style-type: none"> <li>· <b>Transport/Additional information:</b></li> </ul>                                                              | <p>Not dangerous according to the above specifications.</p> |
| <ul style="list-style-type: none"> <li>· <b>UN "Model Regulation":</b></li> </ul>                                                                         | <p>Not Reguated</p>                                         |

### SECTION 15: Regulatory information

- **15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**
- **Poisons Act - Schedule 1**  
None of the ingredients is listed.

(Contd. on page 7)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 12.06.2020

**Trade name: Bio-Safe Coomassie Stain**

(Contd. of page 6)

**· Poisons Act - Schedule 2, Group II**

None of the ingredients is listed.

**· Health Products Act - First Schedule - Psychotropic Substances**

None of the ingredients is listed.

**· Directive 2012/18/EU**
**· Named dangerous substances - ANNEX I** None of the ingredients is listed.

**· National regulations**
**· Technical instructions (air):**

| Class | Share in % |
|-------|------------|
| I     | 0.1-1      |
| NK    | 0.01-0.1   |

**· Water hazard class:** Water hazard class 1 (Self-assessment): slightly hazardous for water.

**· 15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

### SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

**· Department issuing SDS:** Environmental Health and Safety.

**· Contact:**

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

**· Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Flam. Liq. 2: Flammable liquids – Category 2

Acute Tox. 3: Acute toxicity - oral – Category 3

Acute Tox. 5: Acute toxicity - inhalation – Category 5

Skin Corr. 1B: Skin corrosion/irritation – Category 1B

Skin Corr. 3: Skin corrosion/irritation – Category 3

STOT SE 1: Specific target organ toxicity (single exposure) – Category 1

**· \* Data compared to the previous version altered.**

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

### SECTION 1: Identification

- **1.1 Product identifier**
- **Trade name:** Dithiothreitol (DTT)
- **Article number:** 1610610, 1610611, 1610610EDU, 1610611EDU
- **CAS Number:**  
27565-41-9
- **EC number:**  
248-531-9
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**  
No further relevant information available.
- **Application of the substance / the mixture** Laboratory chemicals
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**  
Bio-Rad Laboratories Pte Ltd.  
27 International Business Park 01-02  
Singapore 609924
- phone: 65-6415-3188 fax: 65-6415-3189
- **Information department:** sales.singapore@bio-rad.com
- **1.4 Emergency telephone number:** 65-6415-3188

### SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**  
Acute Tox. 4 H302 Harmful if swallowed.  
Skin Irrit. 2 H315 Causes skin irritation.  
Eye Irrit. 2A H319 Causes serious eye irritation.  
STOT SE 3 H335 May cause respiratory irritation.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No 1272/2008**  
The substance is classified and labelled according to the CLP regulation.
- **Hazard pictograms**



GHS07

- **Signal word** Warning
- **Hazard-determining components of labelling:**  
(R\*,R\*)-(±)-1,4-dimercaptobutane-2,3-diol
- **Hazard statements**  
H302 Harmful if swallowed.  
H315 Causes skin irritation.  
H319 Causes serious eye irritation.  
H335 May cause respiratory irritation.
- **Precautionary statements**  
P261 Avoid breathing dust/fume/gas/mist/vapours/spray.

(Contd. on page 2)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

Trade name: Dithiothreitol (DTT)

(Contd. of page 1)

- P280 Wear protective gloves / eye protection / face protection.  
 P301+P312 IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.  
 P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
 P405 Store locked up.  
 P501 Dispose of contents/container in accordance with local/regional/national/international regulations.
- **2.3 Other hazards**
  - **Results of PBT and vPvB assessment**
  - **PBT:** Not applicable.
  - **vPvB:** Not applicable.

### SECTION 3: Composition/information on ingredients

- **3.1 Chemical characterisation: Substances**
- **CAS No. Description:**  
27565-41-9 (R\*,R\*)-(±)-1,4-dimercaptobutane-2,3-diol
- **Identification number(s):**
- **EC number:** 248-531-9
- **Additonal information:** For the wording of the listed risk phrases refer to section 16.

### SECTION 4: First-aid measures

- **4.1 Description of first aid measures**
- **General information**  
Immediately remove any clothing soiled by the product.  
Symptoms of poisoning may even occur after several hours; therefore medical observation for at least 48 hours after the accident.
- **After inhalation** Supply fresh air; consult doctor in case of complaints.
- **After skin contact**  
Immediately wash with water and soap and rinse thoroughly.  
Immediately rinse with water.
- **After eye contact** Rinse opened eye for several minutes under running water. If symptoms persist, consult a doctor.
- **After swallowing**  
Call for a doctor immediately.  
Rinse mouth with water. Seek medical attention and appropriate follow-up.
- **4.2 Most important symptoms and effects, both acute and delayed**  
Nausea  
Disorientation
- **4.3 Indication of any immediate medical attention and special treatment needed**  
No further relevant information available.

### SECTION 5: Fire-fighting measures

- **5.1 Extinguishing media**
- **Suitable extinguishing agents**  
Use fire fighting measures that suit the environment.

(Contd. on page 3)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

Trade name: Dithiothreitol (DTT)

(Contd. of page 2)

- CO<sub>2</sub>, extinguishing powder or water spray. Fight larger fires with water spray or alcohol resistant foam.
- **5.2 Special hazards arising from the substance or mixture** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** Wear self-contained respiratory protective device.

### SECTION 6: Accidental release measures

- **6.1 Personal precautions, protective equipment and emergency procedures** Wear protective clothing.
- **6.2 Environmental precautions:** Do not allow to enter sewers/ surface or ground water.
- **6.3 Methods and material for containment and cleaning up:**  
Dispose contaminated material as waste according to item 13.  
Ensure adequate ventilation.
- **6.4 Reference to other sections**  
See Section 7 for information on safe handling  
See Section 8 for information on personal protection equipment.  
See Section 13 for disposal information.

### SECTION 7: Handling and storage

- **7.1 Precautions for safe handling** Ensure good ventilation/exhaustion at the workplace.
- **Information about protection against explosions and fires:** No special measures required.
- **7.2 Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and receptacles:** Store in a cool location.
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** Store in cool, dry conditions in well sealed receptacles.
- **7.3 Specific end use(s)** No further relevant information available.

### SECTION 8: Exposure controls/personal protection

- **Additional information about design of technical systems:** No further data; see item 7.
- **8.1 Control parameters**
- **Components with limit values that require monitoring at the workplace:** Not required.
- **Additional information:** The lists that were valid during the creation were used as basis.
- **8.2 Exposure controls**
- **Personal protective equipment**
- **General protective and hygienic measures**  
Keep away from foodstuffs, beverages and feed.  
Immediately remove all soiled and contaminated clothing  
Wash hands before breaks and at the end of work.  
Avoid contact with the eyes.  
Avoid contact with the eyes and skin.
- **Breathing equipment:**  
Filter P1.

(Contd. on page 4)



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

Trade name: Dithiothreitol (DTT)

(Contd. of page 3)

In case of brief exposure or low pollution use respiratory filter device. In case of intensive or longer exposure use respiratory protective device that is independent of circulating air.

· **Protection of hands:** Protective gloves.

· **Material of gloves**

Recommended thickness of the material:  $\geq 0.11$  mm

Nitrile rubber, NBR

Synthetic gloves

· **Penetration time of glove material**

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

· **For the permanent contact gloves made of the following materials are suitable:** Nitrile rubber, NBR

· **As protection from splashes gloves made of the following materials are suitable:** Nitrile rubber, NBR

· **Eye protection:**

Safety glasses

Tightly sealed goggles.

### SECTION 9: Physical and chemical properties

#### · 9.1 Information on basic physical and chemical properties

##### · General Information

##### · Appearance:

Form: Solid.

Colour: Whitish

· Odour: Sulfurous

· Odour threshold: Not determined.

· pH-value: Not applicable.

##### · Change in condition

Melting point/freezing point: 42-44 °C

Initial boiling point and boiling range: undetermined

· Flash point: 110 °C

· Flammability (solid, gaseous) Product is not flammable.

· Decomposition temperature: Not determined.

· Self igniting: Not determined.

· Explosive properties: Product does not present an explosion hazard.  
Not determined.

##### · Explosion limits:

Lower: Not determined.

Upper: Not determined.

· Vapour pressure: Not applicable.

· Density: Not determined

· Relative density: Not determined.

· Vapour density: Not applicable.

· Evaporation rate: Not applicable.

(Contd. on page 5)



## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

Trade name: Dithiothreitol (DTT)

(Contd. of page 4)

- |                                                           |                                            |
|-----------------------------------------------------------|--------------------------------------------|
| · <b>Solubility in / Miscibility with Water at 25 °C:</b> | 0.5 g/l                                    |
| · <b>Partition coefficient: n-octanol/water:</b>          | Not determined.                            |
| · <b>Viscosity:</b>                                       |                                            |
| dynamic:                                                  | Not applicable.                            |
| kinematic:                                                | Not applicable.                            |
| · <b>Solids content:</b>                                  | 100.0 %                                    |
| · <b>9.2 Other information</b>                            | No further relevant information available. |

### SECTION 10: Stability and reactivity

- **10.1 Reactivity** No further relevant information available.
- **10.2 Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **10.3 Possibility of hazardous reactions** No dangerous reactions known
- **10.4 Conditions to avoid** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known

### SECTION 11: Toxicological information

- **11.1 Information on toxicological effects**

- **Acute toxicity**

Harmful if swallowed.

- **LD/LC50 values that are relevant for classification:**

**27565-41-9 (R\*,R\*)-(±)-1,4-dimercaptobutane-2,3-diol**

Oral LD50 400 mg/kg (rat)

- **Primary irritant effect:**

- **Skin corrosion/irritation**

Causes skin irritation.

- **Serious eye damage/irritation**

Causes serious eye irritation.

- **Respiratory or skin sensitisation** Based on available data, the classification criteria are not met.

- **CMR effects (carcinogenity, mutagenicity and toxicity for reproduction)**

- **Germ cell mutagenicity** Based on available data, the classification criteria are not met.

- **Carcinogenicity** Based on available data, the classification criteria are not met.

- **Reproductive toxicity** Based on available data, the classification criteria are not met.

- **STOT-single exposure**

May cause respiratory irritation.

- **STOT-repeated exposure** Based on available data, the classification criteria are not met.

- **Aspiration hazard** Based on available data, the classification criteria are not met.

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(Contd. on page 6)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

Trade name: Dithiothreitol (DTT)

(Contd. of page 5)

### SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability** No further relevant information available.
- **12.3 Bioaccumulative potential** No further relevant information available.
- **12.4 Mobility in soil** No further relevant information available.
- **Additional ecological information:**
- **General notes:**  
Water hazard class 1 (German Regulation) (Self-assessment): slightly hazardous for water.  
Do not allow undiluted product or large quantities of it to reach ground water, water course or sewage system.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **12.6 Other adverse effects** No further relevant information available.

### SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation** Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.
- **Recommended cleansing agent:** Water, if necessary with cleansing agents.

### SECTION 14: Transport information

- |                                                                                                                                                           |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <ul style="list-style-type: none"> <li>· <b>14.1 UN-Number</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> </ul>                                          | Not Reguated    |
| <ul style="list-style-type: none"> <li>· <b>14.2 UN proper shipping name</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> </ul>                            | Not Reguated    |
| <ul style="list-style-type: none"> <li>· <b>14.3 Transport hazard class(es)</b></li> <li>· <b>ADR, ADN, IMDG, IATA</b></li> <li>· <b>Class</b></li> </ul> | Not Reguated    |
| <ul style="list-style-type: none"> <li>· <b>14.4 Packing group</b></li> <li>· <b>ADR, IMDG, IATA</b></li> </ul>                                           | Not Reguated    |
| <ul style="list-style-type: none"> <li>· <b>14.5 Environmental hazards:</b></li> </ul>                                                                    | Not applicable. |
| <ul style="list-style-type: none"> <li>· <b>14.6 Special precautions for user</b></li> </ul>                                                              | Not applicable. |
| <ul style="list-style-type: none"> <li>· <b>14.7 Transport in bulk according to Annex II of Marpol and the IBC Code</b></li> </ul>                        | Not applicable. |

(Contd. on page 7)

## Safety data sheet according SS 586 : Part 3

Printing date 25.08.2020

Revision: 30.07.2020

Trade name: Dithiothreitol (DTT)

(Contd. of page 6)

|                          |              |
|--------------------------|--------------|
| · UN "Model Regulation": | Not Reguated |
|--------------------------|--------------|

### SECTION 15: Regulatory information

#### · 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

##### · Poisons Act - Schedule 1

Substance is not listed.

##### · Poisons Act - Schedule 2, Group II

Substance is not listed.

##### · Health Products Act - First Schedule - Psychotropic Substances

Substance is not listed.

##### · Directive 2012/18/EU

· **Named dangerous substances - ANNEX I** Substance is not listed.

##### · National regulations

· **Water hazard class:** Water hazard class 1 (Self-assessment): slightly hazardous for water.

· **15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

### SECTION 16: Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

· **Department issuing SDS:** Environmental Health and Safety.

#### · Contact:

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

#### · Abbreviations and acronyms:

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Acute Tox. 4: Acute toxicity - oral - Category 4

Skin Irrit. 2: Skin corrosion/irritation - Category 2

Eye Irrit. 2A: Serious eye damage/eye irritation - Category 2A

STOT SE 3: Specific target organ toxicity (single exposure) - Category 3

· **\* Data compared to the previous version altered.**